RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR

Department

of

Computer Science Engineering

Data Mining Lab

Using

Advance Python Programming

For

B.Tech-CSE-E3-Sem-2

Academic Year: 2019-20

DATAMINING Using PYTHON

NOTE:

Make sure that students must and should have basic idea above Python Programming:

Download the required Dataset

Software Required: Anaconda 3 or Jupyter

Contents

Laboratory Experiment List

Week1:

Task: Data Wrangling

Required Python Libraries: numpy, pandas

Week2:

Task: DataPre-processing and visualization

Required Python Libraries: pandas, matplotlib, seaborn

Week3:

Task: Association Rule Mining and Simple Linear Regression

Required Python Libraries: pandas, matplotlib, numpy

Week4:

Task: Multiple Linear Regression

Required Python Libraries: pandas, matplotlib, numpy

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Week5:

Task: Decision Tree

Required Python Libraries: pandas, matplotlib, seaborn

Week6:

Task: Navie Bayes

Required Python Libraries: pandas, matplotlib, numpy

Week7:

Task: K Nearest Neighbor KNN

Required Python Libraries: pandas, matplotlib, numpy

Week8:

Task: Support Vector Machine-Classification of IRIS Flowers

Required Python Libraries: pandas, matplotlib, numpy, seaborn

Week9:

Task: Artificial Neural Network

Required Python Libraries: numpy, pandas

Week10:

Task: K-Means Clustering

Required Python Libraries: pandas, matplotlib, numpy

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Detailed Problem Solving :

Week1:

Task: Data Wrangling

Required Python Libraries: numpy, pandas

Problems:

Import Dataset [transactions.csv]

1. Print Summary of transaction data set
2. Print Numbers of Attributes
3. Print Numbers of Records
4. Get the row names
5. Get the column names
6. View top 10 Records
7. Change the name of column " Quantity" to "Quant"
8. Change the name of columns ProductID and UserID to PID and UID respectively
9. Order the rows of transactions by TransactionId descending, if ascending then ascending=True
10. Order the rows of transactions by Quantity ascending, TransactionDate descending
11. Set the column order of Transactions as ProductID, Quantity, TransactionDate, TransactionID, UserID
12. Make UserID the first column of transactions
13. Extracting arrays from a Data Frame. Get the 2nd column
14. Get the ProductID Array
15. Get the ProductId array using the following variable

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16.Row subsetting, subset rows 1,3 and 6
17. subset rows excluding 1,3 and 6
18. Subset the first three rows
19. Subset the last 2 rows
20. Subset rows excluding the last 2 rows
21. Subset rows excluding the first 3 rows
22. Subset rows where Quantity>1
23. Subset rows where UserID=2
24. Subset rows where Quantity>1 and UserID=2
25. Subset rows where Quantity + UserID is >3
26. Subset rows where an external array, foo, is True
27. Subset rows where an external array, bar, is positive
28. Subset rows where foo is TRUE or bar is negative
29. Subset the rows where foo is not TRUE and bar is not negative
30. Subset by columns 1 and 3
31. Subset by columns TransactionID and TransactionDate
32. Subset by columns TransactionID and TransactionDate with logical operator
33. Subset columns by a variable list of column names
34. Subset columns excluding a variable list of column names
35. Inserting and updating values
36. Convert the TransactionDate column to type Date
37. Insert a new column, Foo=UserID+ProductID

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Week2:

Task: DataPre-processing and visualization

Required Python Libraries: pandas, matplotlib, seaborn

Problems:

Import Dataset [car_sales.csv]

1. Read top five values
2. Print dataframe info_data types of each column
3. Print number of rows and columns
4. Drop duplicate rows_if any
5. Print number of rows and columns after dropping duplicates
6. Print summary statistics for numerical variables
7. Print number of missing values in each column
8. Drop the column with most missing values
9. Drop the rows with categorical missing values
10. Import the rows with numerical missing values
11. Sort the data w.r.t price_find the details of the most and the least expensive cars
12. Write a function to find min and max values of any column
13. Call the above function to find min max of horse power, length, fuel efficiency
14. Plot histogram of continuous numerical variable : price,sales,hoursepower,fuelefficiency
15. Probability density distribution of continuous numerical variable-length

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16. Count by category –group by manufacture
17. Select all numerical variables
18. Print correlation coefficient value of price and sales
19. Plot correlation of price and sales using scatterplot
20.Pair plot
21. Boxplot of sales of different manufacturer
22. Boxplot of other numerical variables w.r.t manufacture
23. Divide the data into input and output y=sales in thousand, x=all other variables
24. Encode other categorical variables using label encoder
25. Encode categorical variable vehicle type using one-hot encoder
26.Split the data set into train and test set 70% train set 10% test set
27. Apply feature scaling on numerical variables

Week3:

Task: Association Rule Mining and Simple Linear Regression

Required Python Libraries: pandas, matplotlib, numpy

1. Import libraries
2. Data preprocessing
3. Apply association rules mining on the groceries dataset

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4. Use min-sup=0.003, min-cont=0.3, min-lift=2 and min-length=2
5. Training apriori on the dataset
6. Visualizing the results
7. Play with min-sup to find the rule with maximum support value (keeping other configuration same)
8. Play with min-cont to find the rule with maximum confidence value (keeping other configurations same)
9. Play with min-lift to find the rule with maximum lift value (keeping other configurations same)
10. Import necessary libraries
11. Import dataset and display first 5 rows
12. Display last 5 rows
13. Check column data types
14. Check the shape of the data Frame (rows,columns)
15. Print mean squared error and R squared value
16. Calculate and print R squared and adjusted R squared values without any library

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Week4:

Task: Multiple Linear Regression

Required Python Libraries: pandas, matplotlib, numpy

1. Import necessary libraries
2. Import the dataset and look at the top five rows

3. Look at the bottom five rows
4. Check the shape of the data frame(rows,columns)
5. Look at the data types of the columns
6. Look at the summary statistics
7. Do a pairplot of the variables
8. Visualize the relationship between the independent variables and the dependent variable using pairplot
9. Prepare X(independent variables) and Y (dependent variable)
10. Split data into training and testing sets (70%-train,30%-test)-X-train ,Y-train, X-test, Y-test
11. Fitting multiple linear regression to the training set
12. Predict on the train set and calculate the error=Y-pred-Y-train
13. Residual plot-plot y-pred in x axis and errors in y axis
14. Product on the test set
15. Print mean squared error and R squared value
16. Build the optional mode using backward elimination
17. Import the dataset (housing.csv)&look at the top five rows
18. Look at the bottom five rows
19. Check the shape of the data frame(rows,columns)
20. Look at the data type of the columns

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21. Look at the summary statistics
22. Prepare X(independent variable) and Y (dependent variable)
23. Encoding categorical data in X
24. Avoiding the dummy variable trap
25. Look at the top 5rows after preprocessing
26. Apply feature scaling on numerical variables
27. Split the data into training and testing seta (70%Train,30%Test)-X-Train,X-test,Y-train,
28. Backward elimination Build the optimal model using backward elimination

Week5:

Task: Decision Tree

Required Python Libraries: pandas, matplotlib, seaborn

1. Import the necessary libraries
2. Load the data sets train and test
3. Look at top values of train data
4. Print the column names
5. Check the data type of each
6. Print the dimension of train and test data
7. Print summary statistics of train data
8. Print the no.of missing values in each column of train and test data
9. Barchart of survived, pclass, sex, embarked

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10. Plot histogram of age for survived and did not survived class
11. Replace the mission values in Embarked column by ‘S’
12. Impute the missing value in fare column wit median value
13. Impute the age column. It has too many missing values but it may be import
14. That has same mean and standard deviation as non missing age data
15. Feature engineering creating new features Create new feature familysize as a combination of sibsp and parch
16. Family size=parch+sibsp+1
17. Create new feature isalone from family size
18. If family size==1 then isalone =True else isalone=false
19. Drop the columns which may not be useful in prediction of survival
20. Drop passenger ID,Name.Ticket and Cabin
21. Print the top five elements
22. Do a pairpolt of all features
23. Prepare X and Y
24. Splitting the data into train and validation set
25. Create th model

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26. Fitting the decision tree with default hyper parameter
27. Print the tree
28. Check the evaluation metrics of our default mode confusion matrix accurary
29. Importing classification report and confusion matrix from sklearn metrics
30. Macking predictions
31. Printing classification report
32. Printing confusion matrix and accuracy
33. Grid search CV to find optimal max-dept
34. Specfy no.of folds for K-flod CV
35. Parameters to buld the model on
36. Instantiate the model
37. First tre on training data
38. Score of grid search CV
39. Plotting accuracies with max-depth
40. Create a model with optimal hyper parameters max-depthg, min-samples-leaf, min-samples-split
41. Decide the optimal values by looking at the previous plotting
42. Print the optimal model
43. Lets check the evaluation metrics of our optimal model
44. Importing classification report and confusion matrix from sklearn metrics

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45. Making predections
46. Printing classification report
47. Print confusion matrix and accuracy

Week6:

Task: Navie Bayes

Required Python Libraries: pandas, matplotlib, numpy

1. Import required libraries
2. Import 20news group dataset from scikit learn datasets
3. Load 20news group train subset
4. Load 20 news group test subset
5. Print all target labels
6. Prepare subset of categories alt.athesssm com.graphics sci.space
7. Load 20 news group train subset with three categories alt.atheism; comp.graphic
8. Load 20 news group test subset with three categories alt.atheism; com.graphics;
9. Print new training set target names(Labels)
10. Print news training data of 5th article
11. Print shape of data and targets
12. Print training set filenames
13. By using Count vectorizer train data into numerical format considering
14. Use bernoullNB for training

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15. By using countvectorizer convert test data into numeric format considering only
16. Predict target labels for testing set
17. Find accuracy score on test set
18. Used Tfidfvectorizer instead of count vectorizer and use multinomial NB
19. Find test set accuracy
20. Try with avoiding stopwords and repet the same

Week7:

Task: K Nearest Neighbor KNN

Required Python Libraries: pandas, matplotlib, numpy

1. Import necessary libraries
2. Import the train data
3. Check shape of the data
4. Select only 10%of the data from experiment. We will not use total data as KNN is
5. Print shape of selected dataset
6. Look at the top 5 rows
7. Find no of unique class labels
8. Print the no of examples for each class label
9. Prepare X and Y
10. Visualization the image of the first 5 digital in X- also print the class label in

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11. Split the train dataset into training set(75%)&validation set(25%)
12. Fit KNN classifies to training data
13. Default parameters n-neighbors-5,metric='minkowski' p=2
14. Predict the validation set results
15. Print the confusion matrix
16. Printing classification report and accuracy
17. Visualize the image of the first 5 digits in X-test also print the predicted

Week8:

Task: Support Vector Machine-Classification of IRIS Flowers

Required Python Libraries: pandas, matplotlib, numpy, seaborn

1. Import necessary libraries
2. Load dataset
3. Print no.of rows and columns
4. Print dataframe into data types
5. Print attribute name
6. Print top 5 rows
7. Print bottom 5 rows
8. Print the unique class labels

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9. Print no of examples from each class label print no of missing values in each column
10. Drop the ID column
11. Print summary statics
12. Plot histogram of all input features
13. Create a scatter plot of ‘sepal length cm’vs’sepalwidthcm’
14. Create a scatter plot of ‘sepalalengthcm’vs’sepal widthcm’ using seaborn’s facecrid
15. Boxplot of petalwidth for different species
16. Boxplot of petalLength for different species
17. Pairpolt of all input features colosed by species(Hint : use the hue parameter)
18. Print the correlation among different features
19. Prepare x and y
20. Split into train and test data-70% train random state=1
21. Look at the documentation of the SVC class thet you will use to create the model
22. Model building
23. Instatiate an object of class CSVSC using cost c=1
24. Fit the model
25. Predict using the model
26. Evalution the model using confusion matrix
27. Print other metrics

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28. Print classification report and accuracy
29. K-fold cross validation
30. Use grid search CV to find optimal value of c-sel decision tree lab questions
31. Try with C=[0.1,1,10,100,1000]
32. Specfy no of folds for K-fold CV
33. Fit tree on training data
34. Print the scores of grid search cv as data frame
35. Plot of c verrus train and test scores
36. Print the best score and best C value model CV is the grid search CVobject

Week9:

Task: Artificial Neural Network

Required Python Libraries: numpy, pandas

1. Titanic dataset
2. Import necessary libraries
3. Encoding the categorical features
4. Feature set
5. Back propagation step2
6. Plot the wcss values
7. Fead forward step2
8. Print the confusion matrix

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Week10:

Task: K-Means Clustering

Required Python Libraries: pandas, matplotlib, numpy

1. Import libraries
2. Load the data set
3. Look at top 5 values
4. Select only annual income and spending score
5. Import KMeans clustering
6. List to store within cluster sum of square
7. Fit the KMeans algorithm with different values of K
8. Plot the WCSS values
9. Fit the Kmeans clustering algo with K=5
10. Make predication
11. Visualizing the clusters

***** END *****

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